CSA47 – DEEP LEARNING

ASSESSMENT 2

Analyse the Mathematics of a Diabetic Retinopathy Screening Model

RetinaCare — Mathematical Analysis and Screening Audit

Student detail	Enter your details
Student name	MOHAMED JALAL AG
Register number	192321130
Course	CSA4704 – Deep Learning
Assessment	CO1 Individual assessment
Total marks	50
Working environment	Google Colab / Python

Task 1 – Break down the audit problem (3 marks)

Data item	Given information	Mathematical concept	Model stage
D1	A=[[3,0],[4,5]]	SVD, rank-1 reconstruction	Representation / compression
D2	$J(w)=\frac{1}{2}w^THw-b^Tw$; H=[[4,1],[1,4]], $b=[5,5]^T$	Gradient, Hessian, convexity, condition number, GD, momentum	Loss / optimisation
D3	p=[0.5,0.25,0.125,0.125] and two q distributions	Entropy, cross-entropy, KL divergence	Probability / uncertainty
D4	Prevalence 10%, sensitivity 90%, specificity 95%	Bayes theorem and conditional probability	Decision / screening
D5	True score 10, noise variance 1, model predictions P and Q	Bias, variance, expected squared error	Prediction error
D6	TP/FN/FP/TN counts from 100 images	Accuracy, precision, recall, specificity, F1	Evaluation

These are independent numerical examples; do not assume that all values arise from one jointly trained model. D1 and D2 use linear algebra and optimisation; D3 compares probability distributions; D4 depends on prevalence and diagnostic rates; D5 treats supplied predictions as replicate estimates; D6 treats "referable" as the positive class.

Meaning for the system: The audit combines representation, training, uncertainty, clinical decision-making and evaluation. Keeping each data item separate avoids drawing claims that the problem has not supplied evidence for.

Task 2 – Compressibility of feature matrix (6 marks)

Given A=[[3,0],[4,5]]. The singular values follow from the eigenvalues of A^TA.

A^T = [[3,4],[0,5]]

A^TA = [[3,4],[0,5]][[3,0],[4,5]] = [[25,20],[20,25]]

Each element is a row-by-column product: $3^2+4^2=25$; $3\times0+4\times5=20$; $0\times3+5\times4=20$; $0^2+5^2=25$.

$\det(A^TA-\lambda I) = (25-\lambda)^2-20^2 = 0$

$\lambda^2-50\lambda+225=0$ $\lambda=[50\pm\sqrt{(2500-900)}]/2=[50\pm40]/2$

Therefore $\lambda_1=45.0000$ and $\lambda_2=5.0000$. The right singular vectors are the unit eigenvectors: $v_1=(1/\sqrt{2})[1,1]^T$, $v_2=(1/\sqrt{2})[1,-1]^T$, so $V=(1/\sqrt{2})[[1,1],[1,-1]]$.

$\sigma_i=\sqrt{\lambda_i}$ $\sigma_1=\sqrt{45}=6.7082$; $\sigma_2=\sqrt{5}=2.2361$; $\sigma_1/\sigma_2=3.0000$

Use $u_i=Av_i/\sigma_i$. Then $u_1=(1/\sqrt{10})[1,3]^T$ and $u_2=(1/\sqrt{10})[3,-1]^T$, hence $U=(1/\sqrt{10})[[1,3],[3,-1]]$ and $\Sigma=\text{diag}(6.7082,2.2361)$. Thus $A=U\Sigma V^T$.

Matrix	Approximate form
U	[[0.3162, 0.9487], [0.9487, −0.3162]]
Σ	[[6.7082, 0.0000], [0.0000, 2.2361]]
V^T	[[0.7071, 0.7071], [0.7071, −0.7071]]

$A_1=\sigma_1 u_1 v_1^T=(3/2)[[1,1],[3,3]]=[[1.5000,1.5000],[4.5000,4.5000]]$

$A-A_1=[[1.5000,-1.5000],[-0.5000,0.5000]]$

$||A-A_{\ }||_F=\sqrt{(1.5^2+1.5^2+0.5^2+0.5^2)}=\sqrt{5}=2.2361$

Energy retained = $\sigma_{\ }^2/(\sigma_{\ }^2+\sigma_{\ }^2)\times100\% = 45/(45+5)\times100\% = 90.0000\%$

Meaning for the system: The first singular component carries 90% of the squared matrix energy. Rank-1 is a strong numerical approximation for this small matrix, but actual retinal-image classification must confirm that compressed features preserve clinical signal.

Task 3 – Curvature of the loss surface (5 marks)

Given H=[[4,1],[1,4]] and b=[5,5]^T.

$\det(H-\lambda I)=(4-\lambda)^2-1=0 \quad \lambda_{\ }=5.0000, \lambda_{\ }=3.0000$

For λ=5 the eigenvector has x=y, so $v_{\ }=(1/\sqrt{2})[1,1]^T$. For λ=3 it has x=−y, so $v_{\ }=(1/\sqrt{2})[1,-1]^T$. The first direction has higher curvature.

$\det(H)=4\times4-1\times1=15.0000>0$; eigenvalues are both positive H is positive definite.

Consequently J(w) is strictly convex and has a unique global minimiser. The condition number is $\kappa_2(H)=\lambda_{max}/\lambda_{min}=5/3=1.6667$, a moderate curvature ratio.

$J(w)=Hw-b$; set $J=0$ $w^*=H_{\ }^{-1}b$

$H_{\ }^{-1}=(1/15)[[4,-1],[-1,4]]$; $w^*=(1/15)[[4,-1],[-1,4]][5,5]^T=[1.0000,1.0000]^T$

$J(w^*)=\frac{1}{2}w^{*T}Hw^*-b^Tw^*=\frac{1}{2}(10.0000)-10.0000=-5.0000$

The assessment wording asks for J(w*)=0, but that conflicts with its supplied objective. The minimum of the stated function is −5.0000. Adding +5 to J would move the minimum value to zero without changing the gradient or minimiser.

Meaning for the system: The Hessian is well behaved and gives a unique optimum. A discrepancy in the question's expected loss value should be reported rather than forcing an incorrect answer.

Task 4 – Information theory (5 marks)

Given p=[0.5,0.25,0.125,0.125], $q_{\ }$=[0.25,0.25,0.25,0.25], $q_{\ }$=[0.4,0.3,0.2,0.1]. Use base-2 logarithms.

$H(p)=-\Sigma_i p_i \log_{\ } p_i=0.5000+0.5000+0.3750+0.3750=1.7500$ bits

$H(q)=-\Sigma_i q_i \log_{\ } q_i$

For $q_{\ }$, $H(q_{\ })=-4(0.25\log_{\ }0.25)=2.0000$ bits. For $q_{\ }$, $H(q_{\ })=-[0.4\log_{\ }0.4+0.3\log_{\ }0.3+0.2\log_{\ }0.2+0.1\log_{\ }0.1]=1.8464$ bits.

$H(p,q)=-\Sigma_i p_i \log_{\ } q_i$; $D_{KL}(p||q)=\Sigma_i p_i \log_{\ }(p_i/q_i)=H(p,q)-H(p)$

Quantity (bits)	Untrained q	Partly trained q
H(p)	1.7500	1.7500
H(q)	2.0000	1.8464
H(p,q)	2.0000	1.8007
$D_{KL}(p\|\|q)$	0.2500	0.0507
$D_{KL}(q\|\|p)$	0.2500	0.0536

Verification: for $q_{\ }$, 1.7500+0.2500=2.0000. For $q_{\ }$, 1.7500+0.0507≈1.8007. The partly trained distribution has lower forward KL than the untrained uniform distribution. KL is asymmetric because the weights p_i and q_i differ by direction. Individual class contributions can be negative, while the sum is non-negative. A KL of zero indicates identical distributions (on classes with positive true probability).

Meaning for the system: The probability estimates in $q_{\ }$ are closer to the supplied true distribution than the uniform starting distribution. Cross-entropy minimisation is equivalent to forward-KL minimisation because H(p) stays fixed.

Task 5 – Screening decision using Bayes' rule (4 marks)

Let D mean referable disease. Prevalence P(D)=0.1000; sensitivity P(+|D)=0.9000; specificity P(−|¬D)=0.9500. Thus P(−|D)=0.1000 and P(+|¬D)=0.0500.

Actual condition	Prior	Positive branch	Negative branch
Referable disease	0.1000	0.1000×0.9000=0.0900	0.1000×0.1000=0.0100
No referable disease	0.9000	0.9000×0.0500=0.0450	0.9000×0.9500=0.8550
Total	1.0000	P(+)=0.1350	P(−)=0.8650

P(D|+) = P(+|D)P(D)/P(+) = 0.0900/0.1350 = 0.6667 = 66.6667%

P(D|−) = P(−|D)P(D)/P(−) = 0.0100/0.8650 = 0.0116 = 1.1561%

Sensitivity is P(+|D); PPV is P(D|+). The conditioning is reversed, so they need not match. Among positive tests, 0.0450/0.1350=0.3333 are false positives. A positive screening result should prompt confirmatory assessment, not automatic treatment.

Meaning for the system: At 10% prevalence the positive result implies about a two-thirds probability of disease, and a negative result leaves about 1.16% disease probability under the assumptions.

Task 6 – Gradient-descent convergence (5 marks)

$J(w)=\frac{1}{2}w^{T}Hw-b^{T}w$; $\nabla J(w)=Hw-b$; $w_{k+1}=w_k-\eta(Hw_k-b)$

With $\eta=0.1$ and $w_0=[0,0]^T$:

Step 1: $g_0=[-5,-5]^T$; $w_1=[0,0]-0.1[-5,-5]=[0.5000,0.5000]^T$

$J(w_1)=5(0.5^2)-10(0.5)=-3.7500$

Step 2: $g_1=[-2.5000,-2.5000]^T$; $w_2=[0.7500,0.7500]^T$; $J(w_2)=5(0.75^2)-10(0.75)=-4.6875$

Step 3: $g_2=[-1.2500,-1.2500]^T$; $w_3=[0.8750,0.8750]^T$; $J(w_3)=5(0.875^2)-10(0.875)=-4.9219$

Point	Weights	J(w)	Distance to w*
Initial	[0.0000, 0.0000]	0.0000	1.4142
After step 1	[0.5000, 0.5000]	−3.7500	0.7071
After step 2	[0.7500, 0.7500]	−4.6875	0.3536
After step 3	[0.8750, 0.8750]	−4.9219	0.1768

Distance halves at each step because the iterates lie in the eigen-direction λ=5 and the error multiplier is $1-\eta\lambda=0.5000$. The objective gap to −5 shrinks by a factor of 0.2500 per step in this example.

Meaning for the system: The shrinking distance and objective gap demonstrate convergence at η=0.1. The minimum objective value remains −5, as derived in Task 3.

Task 7 – Momentum and stability (4 marks)

$v_{k+1}=\beta v_k+\nabla J(w_k)$; $w_{k+1}=w_k-\eta v_{k+1}$; β=0.9, η=0.1, $v_0=0$

Iteration 1: $g_0=[-5,-5]$; $v_1=[-5,-5]$; $w_1=[0.5000,0.5000]$; $J(w_1)=-3.7500$

Iteration 2: $g_1=[-2.5,-2.5]$; $v_2=0.9[-5,-5]+[-2.5,-2.5]=[-7,-7]$; $w_2=[1.2000,1.2000]$; $J(w_2)=-4.8000$

Method	Weights after 1	J after 1	Weights after 2	J after 2

Plain GD	[0.5000,0.5000]	−3.7500	[0.7500,0.7500]	−4.6875
Momentum	[0.5000,0.5000]	−3.7500	[1.2000,1.2000]	−4.8000

Momentum overshoots $w^*=[1,1]$ after the second update because part of the previous gradient remains in the velocity. Plain GD on a positive-definite quadratic converges when $0<\eta<2/\lambda_{max}(H)=2/5=0.4000$. With $\eta=0.5$, the multiplier along $\lambda_{max}=5$ is $1-0.5\times5=-1.5000$; its magnitude exceeds one, so error alternates sign and grows.

Meaning for the system: Momentum can accelerate progress in consistent directions but may overshoot. A learning rate beyond the derived bound makes plain GD unstable in the highest-curvature direction.

Task 8 – Prediction-error decomposition (4 marks)

$\text{Bias}^2=(\text{mean prediction}-y)^2$; $\text{Variance}=(1/m)\Sigma\ (\hat{y}\ -\text{mean prediction})^2$; $E[\text{squared error}]\approx\text{Bias}^2+\text{Variance}+\sigma^2$

The supplied target is $y=10$ and noise variance $\sigma^2=1$. For $P=[8,9,8,9]$, mean(P)=8.5000, so $\text{Bias}^2=(8.5-10)^2=2.2500$. The population variance is $((-0.5)^2+(0.5)^2+(-0.5)^2+(0.5)^2)/4=0.2500$. Expected squared error $=2.2500+0.2500+1.0000=3.5000$.

For $Q=[12,7,11,8]$, mean(Q)=9.5000 and $\text{Bias}^2=(9.5-10)^2=0.2500$. Deviations are 2.5, −2.5, 1.5, −1.5; variance=(6.25+6.25+2.25+2.25)/4=4.2500. Expected squared error=0.2500+4.2500+1.0000=5.5000.

Quantity	Model P	Model Q
Mean prediction	8.5000	9.5000
Bias^2	2.2500	0.2500
Variance	0.2500	4.2500
Noise variance	1.0000	1.0000
Expected squared error	3.5000	5.5000

Model P has larger bias but lower variance and is preferred by this decomposition because its expected squared error is 2.0000 lower. This is a decomposition of the given prediction vectors; it is not a substitute for generating a larger bootstrap experiment from actual fitted models.

Meaning for the system: Q estimates the target mean more closely, but its predictions vary much more. The variance dominates Q's error and raises its total expected squared error.

Task 9 – Confusion-matrix metrics (3 marks)

Given TP=40, FN=10, FP=5, TN=45, with N=100.

Accuracy=(TP+TN)/N=(40+45)/100=0.8500; Error rate=(FP+FN)/N=15/100=0.1500

Precision=TP/(TP+FP)=40/45=0.8889; Recall=TP/(TP+FN)=40/50=0.8000

Specificity=TN/(TN+FP)=45/50=0.9000; F1=2TP/(2TP+FP+FN)=80/95=0.8421

Metric	Value	Interpretation
Accuracy	0.8500	85% of all images are classified correctly.
Error rate	0.1500	15% of images are misclassified.
Precision	0.8889	Most positive referrals are true referable cases in this sample.

Recall / sensitivity	0.8000	20% of actual referable cases are missed.
Specificity	0.9000	90% of non-referable cases are correctly excluded.
F1-score	0.8421	Harmonic mean of precision and recall.

Meaning for the system: There are 10 false negatives and 5 false positives. Missed referable cases are more frequent in this sample and have important clinical consequences; accuracy alone hides this distinction.

Task 10 – Relate results to the audit decision (3 marks)

- Compression: the 2×2 matrix rank-1 approximation retains 90.0000% of matrix energy, but it is only a toy example and cannot certify retinal-image compression.
- Optimisation: Hessian eigenvalues 5 and 3 give a positive-definite loss and $\kappa=1.6667$; $\eta=0.1$ converges, while $\eta=0.5$ falls outside the plain-GD stability range.
- Uncertainty and screening: the partly trained q distribution has forward KL 0.0507 bits versus 0.2500 for the uniform q ; Bayes posteriors at 10% prevalence give PPV 0.6667 and disease probability after a negative result 0.0116.
- Clinical error profile: recall is 0.8000, so the supplied confusion matrix misses 10 of 50 actual referable cases. A safe audit must prioritise sensitivity and false-negative consequences as well as accuracy and precision.

Meaning for the system: The mathematical machinery is internally coherent when used with stable calculations and suitable step size, but clinically safe decisions also depend on missed cases, disease prevalence, confirmation pathways and measured performance.

Task 11 – Low-prevalence design challenge (8 marks)

A. Impact analysis (2 marks)

Prevalence changes from 10% to 2%, but the scenario holds sensitivity at 90% and specificity at 95%. These are conditional rates P(+|D) and P(−|¬D), so they do not change solely due to a different population prevalence. Positive predictive value and negative predictive value do change because the prior P(D) enters Bayes' formula.

$$\mathbf{PPV = Se\cdot\pi/[Se\cdot\pi+(1-Sp)(1-\pi)]}$$

$$\mathbf{P(D|-) = (1-Se)\pi/[(1-Se)\pi+Sp(1-\pi)]}$$

B. Posterior probabilities and counts per 1,000 (2 marks)

Prevalence	TP	FN	TN	FP	PPV	P(Disease | Negative)	NPV
10%	90	10	855	45	0.6667	0.0116	0.9884
2%	18	2	931	49	0.2687	0.0021	0.9979

At 10% prevalence, TP=100×0.9=90, FN=100×0.1=10, TN=900×0.95=855, FP=900×0.05=45. PPV=90/(90+45)=0.6667; disease probability after a negative result=10/(10+855)=0.0116.

At 2% prevalence, TP=20×0.9=18, FN=20×0.1=2, TN=980×0.95=931, FP=980×0.05=49. PPV=18/(18+49)=18/67=0.2687; P(D|−)=2/(2+931)=2/933=0.0021 and NPV=931/933=0.9979. The false-referral count increases slightly from 45 to 49, while expected false negatives fall from 10 to 2 because the number of diseased people is much smaller.

C. Policy options and expected cost (2 marks)

Require PPV≥0.5: $Se\pi \geq (1-Sp)(1-\pi)$

$1-Sp \leq Se\pi/(1-\pi)=0.9\times0.02/0.98=0.0183673$

$Sp \geq 1-0.0183673=0.9816327$ minimum specificity = 98.1633%

With sensitivity fixed at 90%, this target yields 18 true positives, 2 false negatives and about 18 false positives per 1,000; PPV is approximately 0.5000. Under the 10:1 cost assumption (10 units per missed case, 1 per unnecessary referral):

Policy scenario	FN per 1,000	FP per 1,000	Cost = 10×FN + FP
10% prevalence, Se=90%, Sp=95%	10	45	145
2% prevalence, Se=90%, Sp=95%	2	49	69
2% prevalence, Se=90%, Sp≈98.1633%	2	≈18	≈38

Raising the positive decision threshold often reduces false positives and raises specificity, but it may reduce sensitivity and increase false negatives. Therefore 98.1633% is a mathematical target under fixed sensitivity, not proof that the score can achieve both rates simultaneously. Choose the operating point from a validation ROC/precision-recall analysis and the stated 10:1 cost.

D. Evidence-based conclusion (2 marks)

Decision: modify the screening policy and validate a prevalence-aware threshold before use. With the current rates at 2% prevalence, PPV is only 0.2687 and there are 49 false referrals and 2 missed cases per 1,000. The higher-specificity target would reduce unnecessary referrals under the fixed-sensitivity assumption, lowering illustrative cost from 69 to about 38 units per 1,000. Because missed disease costs ten times more, the change must be validated without allowing sensitivity to fall below an acceptable clinical minimum. Maintain confirmatory examinations for positive screens and a safety net for persistent symptoms despite negative screens.

Meaning for the system: The original threshold should not be kept without review. A local validation study should select the operating point using the actual prevalence, sensitivity–specificity trade-off, referral capacity and cost of false negatives.

Final numerical answer sheet

Item	Final result
D1 singular values	6.7082 and 2.2361; ratio 3.0000; rank-1 error 2.2361; energy retained 90.0000%
D2 Hessian	Eigenvalues 5.0000, 3.0000; κ=1.6667; $w^*=[1.0000,1.0000]^T$; $J(w^*)=-5.0000$
D3 information theory	H(p)=1.7500; H(q)=2.0000; H(q)=1.8464; CE =1.8007; KL(p||q)=0.0507; KL(q ||p)=0.0536 bits
D4 Bayes at 10%	PPV=0.6667; P(D|−)=0.0116; NPV=0.9884
D5 bias–variance	P total=3.5000; Q total=5.5000; P preferred
D6 metrics	Accuracy 0.8500; error 0.1500; precision 0.8889; recall 0.8000; specificity 0.9000; F1 0.8421
Challenge at 2%	TP=18, FN=2, TN=931, FP=49 per 1,000; PPV=0.2687; NPV=0.9979; cost=69
Specificity target	At least 98.1633% for PPV≥0.5 if sensitivity remains 90%; illustrative cost≈38 units per 1,000

Reproducibility appendix

Run CSA47_Assessment2_RetinaCare_Colab.ipynb in Google Colab to verify SVD, matrix reconstruction error, the Hessian optimum, both KL directions, Bayes posteriors, gradient-descent iterates, momentum updates, bias–variance

values, confusion metrics and the low-prevalence cost table. The notebook output supports the written derivations and does not replace them.